Predicting Postprandial Glycemic Response from Meal Composition and Subject Phenotype

A leave-one-subject-out study on the CGMacros cohort, with external validation on BIG IDEAs

> Research module of the MetaNutri project. This is an independent, reproducible analysis — not a clinical tool, and not wired into the MetaNutri application. Datasets: CGMacros (PhysioNet, DOI `10.13026/3z8q-x658`), licensed **CC BY-NC-SA 4.0**, and BIG IDEAs (PhysioNet, DOI `10.13026/zthx-5212`), licensed **ODC-By 1.0**; see §10 for attribution.

Abstract

Postprandial glucose response (PPGR) varies several-fold between individuals eating the same meal, which makes personalised nutrition attractive but also hard to model. We reproduce and extend the CGMacros baseline on a meal-level table built from 45 subjects and 1,699 continuous-glucose-monitoring (CGM) annotated meals. Using strictly subject-wise **leave-one-subject-out (LOPO)** cross-validation, we predict three response targets — 2-hour incremental area under the curve (iAUC), total area under the curve (AUC), and peak glucose rise — from meal macronutrients plus subject blood-panel and anthropometric features. On the breakfast-only subset our XGBoost model reaches **r = 0.890 for AUC** and **r = 0.655 for iAUC**, matching the published baseline (≈0.89 / ≈0.64), which validates the pipeline. Extending the same features and model to all meal types (1,557 meals) yields **r = 0.838 (AUC)** and **r = 0.451 (iAUC)**. A mean predictor and two macronutrient-only Ridge regressions are far weaker (best linear r = 0.255 for breakfast iAUC), showing that most of the signal is non-linear or subject-specific. We then test cross-cohort generalisation by freezing the CGMacros model and applying it, unchanged, to the independent **BIG IDEAs** cohort (16 subjects, 656 meals, different CGM device and free-living self-reported food logs). AUC transfers partially (**r = 0.569**), but iAUC and peak rise collapse to r ≈ 0.22 — and a simple carbohydrate-only regression transfers *better* (r = 0.36) than the tree ensemble, a cautionary result about non-linear models that fit cohort-specific structure. A within-/between-subject decomposition shows that the high AUC correlation is driven mainly by identifying *who* responds more, whereas iAUC's correlation is closer to genuine meal-level ranking; median-split ROC-AUC is 0.74–0.96 across targets. Distribution-free conformal intervals hit their nominal marginal coverage (0.800 / 0.898) but are wide and, we show candidly, under-cover a minority of individuals. A mixed model with subject random intercepts locates about 69% of AUC variance *between* people versus 32% for iAUC, and shows that, within a person, carbohydrate raises and protein lowers the response. A dedicated reliability-ceiling analysis ties these threads together: the within-cohort model sits below the standardized-breakfast duplicate-meal reliability (r = 0.655 vs ρ = 0.799), iAUC skill is invariant to removing baseline glucose (ruling out circularity), and iAUC transfer fails in *both* cohort directions while AUC transfer is carried almost entirely by baseline glucose (§5). We report the limitations plainly: small cohorts, pooled-observation metrics that mix

between- and within-subject variance, and a large domain gap between the two studies. Code, figures and fixed-seed experiments are included for full reproducibility.

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1. Problem

Two people can eat an identical meal and show very different glucose excursions. Capturing that heterogeneity is the core promise of personalised nutrition, and also its core difficulty: the response depends on meal composition, on the person's metabolic state, and on interactions between the two.

We ask a deliberately narrow, testable question: **given a meal's macronutrients and a set of subject-level clinical features, how well can we predict the 2-hour postprandial glucose response, when the model is evaluated on people it has never seen?** The emphasis on unseen subjects is the point. A model that memorises a person's typical response will look excellent under a random split and fail in deployment, so every number in this report comes from LOPO cross-validation.

The intended use of this work is methodological: it is a writing and engineering sample that demonstrates a correct evaluation protocol on real data. It is not a medical device and makes no diagnostic claim.

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2. Data

2.1 Source and cohort

CGMacros provides CGM traces with time-stamped meal annotations and macronutrient estimates, plus a subject-level `bio.csv` with laboratory measurements. The public release contains 45 subjects. Table 1 summarises the analysis cohort after filtering.

Table 1 — Analysis cohort.

Property	Value
Subjects	45
Meals with iAUC > 0	1,557 (of 1,699 extracted)
Meals per subject (min / median / max)	17 / 35 / 61
Meal types	breakfast 423, lunch 380, dinner 451, snack 303
Age (min / median / max)	18 / 51 / 69
BMI (min / median / max)	20.7 / 30.0 / 49.1
HbA1c % (min / median / max)	4.6 / 5.9 / 8.5
Glycemic status (healthy / prediabetes / T2D)	573 / 576 / 408 meals
Sex (M / F)	16 / 29

The cohort deliberately spans healthy, prediabetic and type-2-diabetic ranges, so the model must handle a wide dynamic range rather than a single narrow population. The dataset is small by machine-learning standards: ~1.5k meals and 45 independent subjects. This shapes every modelling choice below.

2.2 From CGM traces to a meal-level table

For each meal annotation we take the 2-hour postprandial window, sample the glucose series every 15 minutes (9 points), and compute three targets:

- **iAUC** — positive incremental area above the meal-time baseline, using trapezoidal integration with correct handling of curve crossings;
- **AUC** — total trapezoidal area over the window;
- **peak rise** — maximum glucose elevation above baseline.

Features are the four macronutrients (carbohydrate, protein, fat, fiber), the meal-time baseline glucose, and 14 subject-level variables including age, sex, BMI, HbA1c, fasting glucose, insulin, HOMA-IR and a lipid panel (triglycerides, total cholesterol, HDL, non-HDL, LDL, VLDL, cholesterol/HDL ratio). Subject features are joined **on subject id**, not positionally.

Which CGM channel we use, and why it matters. CGMacros participants wore two sensors simultaneously (a Libre Pro sampled every 15 min, and a Dexcom G6 Pro every 5 min). We use the **Libre** channel (`Libre GL`), matching the official baseline, whose corresponding feature is named `Baseline_Libre` — that keeps the replication like-for-like. The choice is not cosmetic: the two channels read systematically differently, and they agree only moderately on *which* meals produce the largest excursions (mean Kendall $\tau \approx 0.43$ across subjects; Howard et al., 2020). Numbers produced on one channel are therefore not directly comparable with numbers produced on the other.

Two data-quality issues were handled explicitly rather than silently. First, column names are inconsistent across the 45 per-subject files (trailing spaces, and one file lacks `Amount Consumed`); we normalise names and access optional columns defensively. Second, meal-type labels are inconsistent (`Snacks`, `snack 1`, `Breakfast`, …); we collapse them to breakfast / lunch / dinner / snack. Only one value is missing across the final feature matrix (a single `fiber_g`), imputed with the **training-fold median**.

Figure 3 shows the resulting data: iAUC distributions differ markedly by meal type, and carbohydrate content correlates only weakly with iAUC ($r \approx 0.2$), foreshadowing the modest ceiling of any meal-composition-only model.

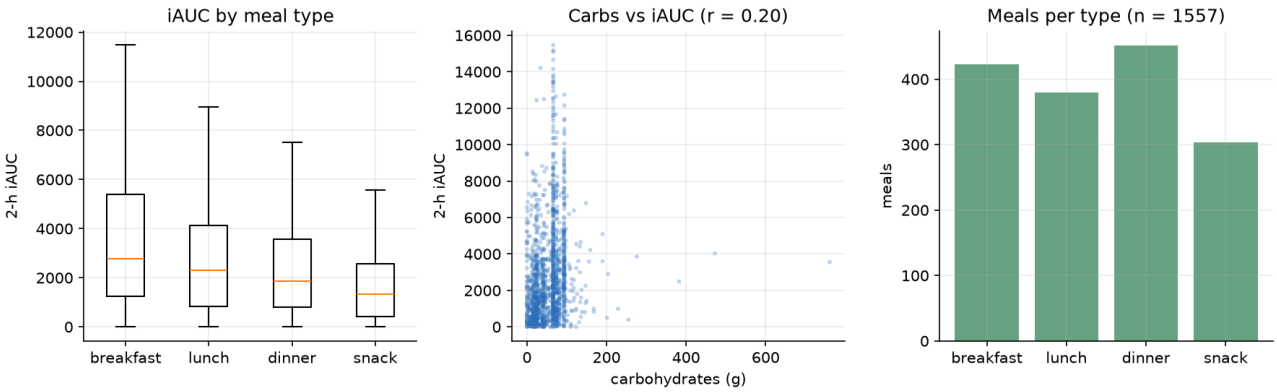


Figure 3 — Data overview. iAUC by meal type; carbohydrate content versus iAUC; meal counts per type.

2.3 External cohort: BIG IDEAs

To test whether the model generalises beyond CGMacros, we use the **BIG IDEAs Lab Glycemic Variability and Wearable Device** dataset as an independent external cohort. It

contributes 16 subjects with Dexcom G6 CGM (5-minute) and a free-living food log. Three differences from CGMacros matter for the analysis:

1. **Meals are not annotated.** The release provides a per-item food log; we group items logged within 20 minutes into one eating event and sum their macronutrients, matching the CGMacros target window (2 h, 9-point sampling).
2. **A narrower feature set.** BIG IDEAs does not record the CGMacros blood panel, so only the five shared features (carbohydrate, protein, meal-time baseline glucose, sex, HbA1c) are used for the primary comparison — complete for all 16 subjects. A seven-feature variant adding fat and fibre covers 13 subjects (three subjects log fat/fibre sparsely or not at all) and is reported as a sensitivity check.
3. **Food-log dates are misaligned in the version we could obtain.** The dataset's own 1.1.3 release notes state "*Updated misaligned food log dates*", confirming the defect. Because 1.1.3 was not retrievable (HTTP 403) and the open mirror carries only 1.1.2, we use 1.1.2 and repair the dates ourselves: for each subject we search integer day offsets and pick the shift that maximises the median post-meal glucose rise, subject to retaining ≥ 90 % of the maximum achievable event coverage. This yields no shift for 12 subjects and a roughly five-month shift for four (007, 013, 015, 016), consistent with the reported defect. As an independent check, the study's standardised-breakfast days show a higher morning glucose rise than other days after correction (median $\Delta \approx +14$ mg/dL).

After correction, the BIG IDEAs table contains **656 meals across 16 subjects**. Per-subject HbA1c ranges from 5.3 % to 6.4 % (median 5.65 %), so the cohort sits in a narrower glycemic range than CGMacros (4.6–8.5 %, including type-2 diabetes); this difference matters when interpreting the AUC transfer asymmetry in §4.4.

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3. Method

3.1 Evaluation protocol

We use **leave-one-subject-out cross-validation**. In each of the 45 folds, all meals from one subject are held out; the model is fit on the remaining 44 subjects and predicts the held-out subject's meals. Standardisation and median imputation are fit on the training fold only, so no information from the test subject leaks into preprocessing. Metrics are computed on the pooled out-of-fold predictions.

This matters quantitatively. Random meal-level splits let the same person appear in train and test, and individual response baselines are highly subject-specific, so random splits substantially overstate performance. Any single-speaker result that is not subject-wise should be read with that optimism in mind.

3.2 Models

We compare four models, deliberately spanning a naive floor and a strong tree-based learner:

1. **Mean predictor** — predicts the training-fold mean. This is the honest floor: anything that cannot beat it has learned nothing.
2. **Carb-only Ridge** — linear regression on carbohydrate.

3. **Energy-only Ridge** — linear regression on carbohydrate, protein and fat.
4. **XGBoost** — gradient-boosted trees with the official CGMacros baseline hyper-parameters: `max_depth = 1`, `n_estimators = 80`, `learning_rate = 0.2`, `reg_alpha = 1.0`.

Ridge, not ordinary least squares, is used for the linear models (`alpha = 1.0`). With 19 strongly collinear features and subjects carrying extreme values, unregularised OLS extrapolates catastrophically on held-out subjects (we observed R^2 below −1,000); a small L2 penalty makes the baseline comparison meaningful rather than a numerical artefact. The tree depth of 1 makes each boosted round a decision *stump*, which is what keeps the model stable in this small-sample regime.

3.3 Metrics

Our primary metric is the **Pearson correlation r** between observed and predicted values under LOPO, which is what the CGMacros baseline reports and therefore what makes replication directly comparable. We additionally report Spearman ρ, R^2, RMSE and MAE. R^2 and RMSE are on the original, untransformed scale (iAUC in mg/dL·min), and are more sensitive to the long right tail than **r** is; we therefore treat **r** as the headline and the error metrics as context. The tree ensemble does not natively produce calibrated uncertainty, so we add distribution-free **conformal prediction intervals** on top of it (§4.6).

Because a pooled correlation conflates two different sources of variance, we additionally report a **within- / between-subject decomposition** (§4.5), and — so the numbers can be compared with classification-style PPGR reporting — a **median-split ROC-AUC** computed from the continuous predictions. The split threshold is derived from the observed responses and is never seen by the model; full per-model values are in `experiments/results.csv`. Repeated meals are handled explicitly — subject random intercept plus within-subject centring, with the within/between effects reported separately — by the mixed model in §4.7.

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4. Results

4.1 Breakfast: replication of the published baseline

Restricting to breakfast (423 meals, 45 subjects) reproduces the official protocol. Table 2 and Figure 2 show the outcome.

Table 2 — Breakfast subset (n = 423, LOPO).

Model	AUC r	iAUC r	Peak r	iAUC R^2	iAUC RMSE
Mean predictor	−0.924	−0.820	−0.826	−0.03	3,474
Carb-only Ridge	−0.014	0.107	0.079	0.00	3,414
Energy-only Ridge	0.100	0.255	0.247	0.06	3,313
XGBoost (official)	**0.890**	**0.655**	**0.632**	**0.43**	**2,589**

Our XGBoost result of **r = 0.890 (AUC)** and **r = 0.655 (iAUC)** aligns with the published ≈0.89 and ≈0.64, which is the main validation that our dataset construction, target definitions and evaluation split are correct. The agreement is close enough that we treat the pipeline as a faithful reproduction, not merely a similar-looking re-run.

The linear baselines expose the structure of the problem. Carbohydrate alone barely correlates with iAUC (r = 0.107); adding protein and fat helps (r = 0.255), consistent with fat and protein modulating gastric emptying and the glycemic curve. But even the best linear model captures only ~6 % of iAUC variance, versus ~43 % for XGBoost. The gap is the empirical case for a non-linear model here.

The mean predictor's strongly negative correlation is a useful diagnostic rather than a real result. Because each fold's prediction is the mean of the *remaining* subjects, a subject with an unusually high response pulls that mean down when held out — inducing an anti-correlation. It confirms that between- subject differences dominate the pooled variance, and it is exactly why a subject-wise split is mandatory.

4.2 All meals: an extension

We then applied the identical features and model to all meal types (1,557 meals), removing the breakfast restriction. Table 3 and Figure 1 summarise.

Table 3 — All-meals subset (n = 1,557, LOPO).

Model	AUC r	iAUC r	Peak r	iAUC R^2	iAUC RMSE
Mean predictor	−0.780	−0.553	−0.552	−0.01	2,735
Carb-only Ridge	−0.032	0.147	0.150	0.02	2,692
Energy-only Ridge	−0.035	0.143	0.146	0.01	2,697
XGBoost (official)	**0.838**	**0.451**	**0.542**	**0.20**	**2,431**

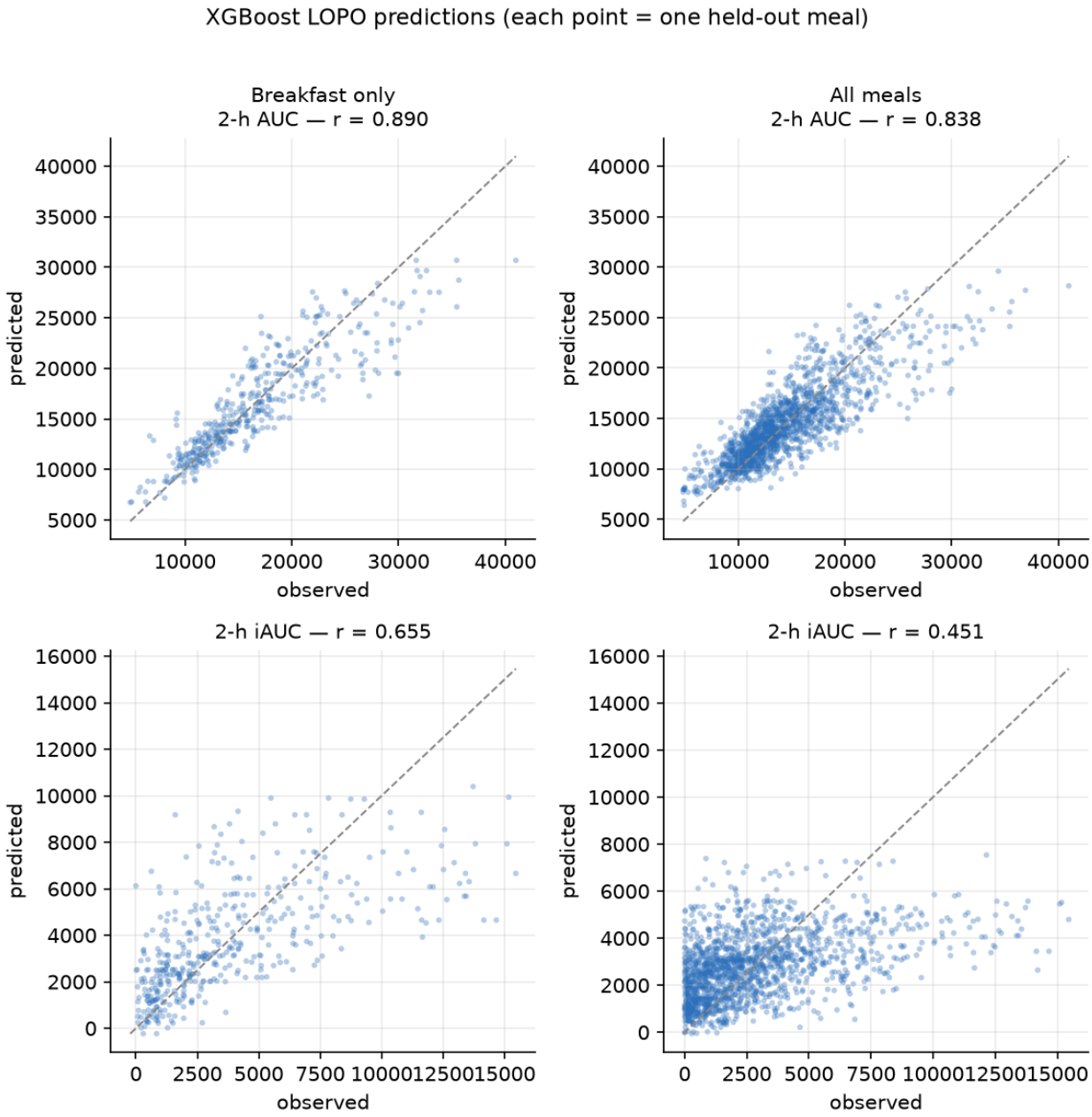


Figure 1 — Predicted versus observed under LOPO. Each point is one held-out meal. Left column: breakfast only; right column: all meals.

AUC remains well predicted (r = 0.838) across meal types. iAUC drops to r = 0.451 and peak rise to r = 0.542. This is the expected direction: breakfasts are the most standardised meal, while lunches, dinners and snacks vary far more in composition, timing and context, and the pooled iAUC distribution becomes heavier-tailed. The model generalises across meal types, but with a clear and honest degradation.

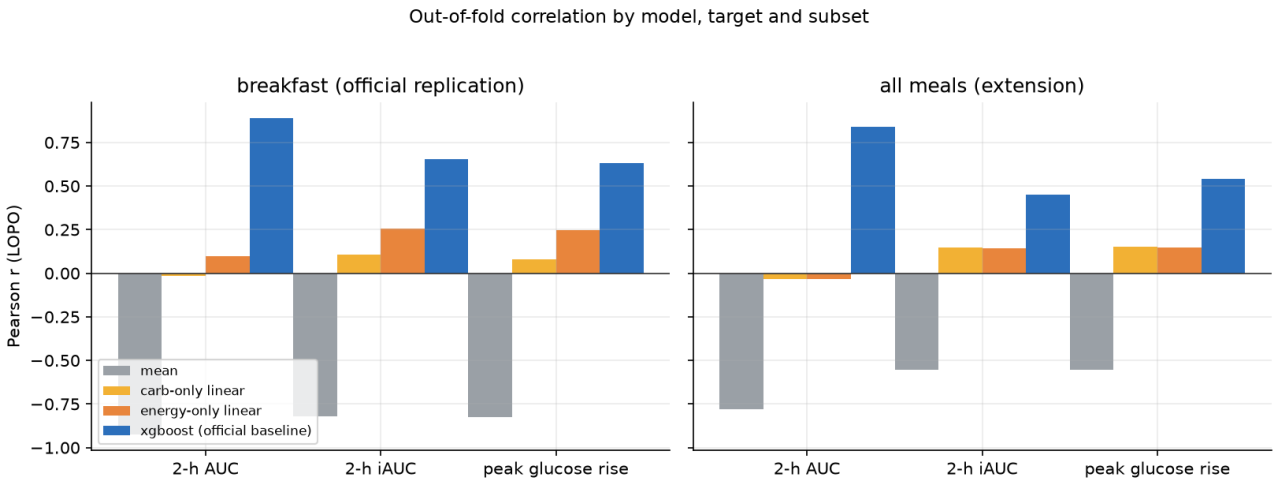


Figure 2 — Model comparison. Out-of-fold Pearson r by model, target and subset. XGBoost dominates everywhere; the linear baselines are close to zero, and the mean predictor is negative by construction. The sign is worth explaining: under LOPO the "mean" model predicts the mean of the *other* 44 subjects, so for a high-responding person that mean is pulled **down** by that person's own exclusion. The leave-one-out mean is therefore anti-correlated with the held-out subject, which is why this no-skill reference lands below zero (and below 0.5 ROC-AUC) instead of at zero. It is a reference point, not a model that has learned anything.

4.3 What the model uses

Figure 4 ranks XGBoost gain importance for iAUC. The dominant features are meal carbohydrate, the meal-time **baseline glucose**, and the subject's glycemic state (HbA1c, fasting glucose). This ordering is physiologically sensible and reassuring: the model leans on the current glucose level and long-run glycemic control, not on incidental lipid variables. Fiber and the lipid panel contribute little once these are present.

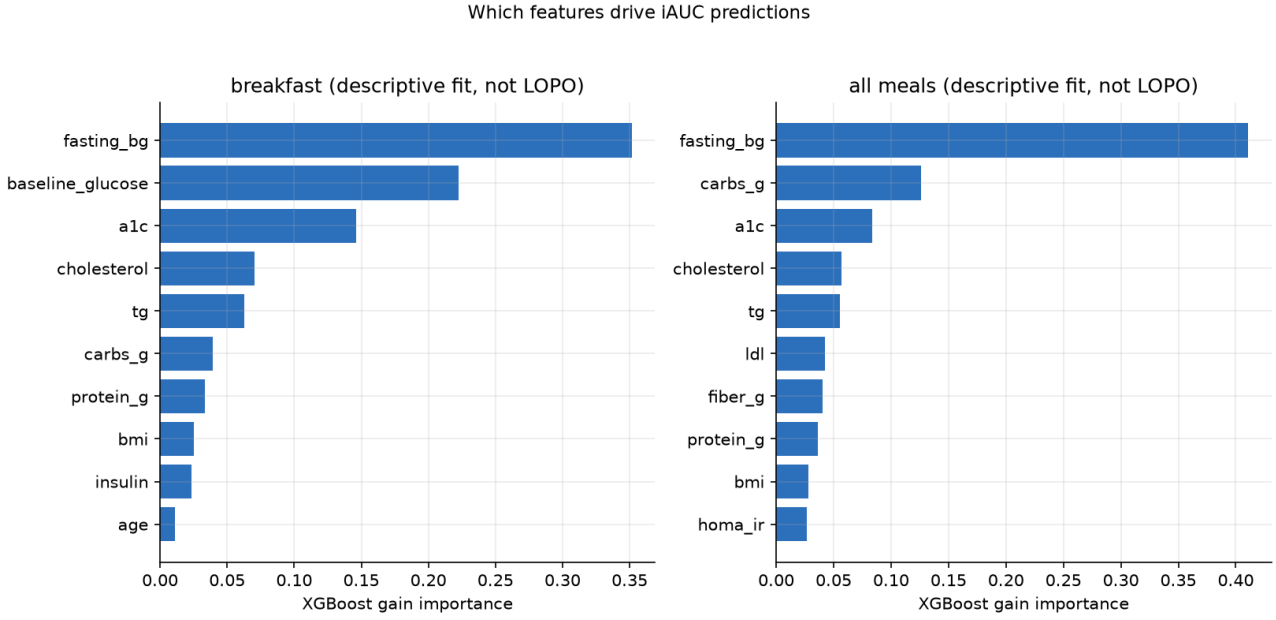


Figure 4 — Feature importance for iAUC. Descriptive fit on all available data (not LOPO); shown for interpretation, never used for model selection.

4.4 External validation on BIG IDEAs

The frozen CGMacros model (same 80 depth-1 trees, same five shared features) is applied to the 656 BIG IDEAs meals without any retraining or refitting of the preprocessing beyond the

CGMacros-fitted scaler. We report three quantities side by side: the model's **internal** LOPO score on each cohort, and its **external transfer** score. Table 4 and Figure 5 show the outcome.

Table 4 — XGBoost Pearson r, 5 shared features.

Cohort / protocol	iAUC r	AUC r	Peak r
CGMacros (LOPO, internal)	0.487	0.816	0.508
BIG IDEAs (LOPO, internal)	0.463	0.625	0.483
BIG IDEAs (external transfer)	**0.227**	**0.569**	**0.223**
BIG IDEAs (external, carb-only Ridge)	0.362	0.288	0.357
BIG IDEAs (external, mean predictor)	— ($R^2 < 0$)	— ($R^2 \approx 0$)	— ($R^2 < 0$)

> Note on sample size: the CGMacros internal row here keeps **all 1,699 meals**, whereas the headline results in §4.2 apply the official replication's `iauc > 0` filter (1,557 meals). The two "internal" CGMacros numbers therefore differ slightly (e.g. all-meal iAUC r 0.487 here vs. 0.451 in §4.2) and should not be compared directly; the BIG IDEAs panels likewise use all 656 meals. See `external_validate.py` for the rationale.

Two things stand out. First, **AUC transfers far better than iAUC**. A correlation of 0.569 for AUC across studies that differ in participants, device and meal annotation is a genuine, if modest, positive result: part of the AUC signal is a portable, physiology-level relationship (glucose level and long-run control). Second, **iAUC and peak rise do not transfer**: r falls from 0.487 and 0.508 internally to 0.227 and 0.223 externally, and R^2 is near zero. The model has learned CGMacros-specific structure for the *excursion* that does not generalise to free-living meals in another cohort.

The internal BIG IDEAs LOPO column is the important control. It shows that BIG IDEAs itself is *not* unlearnable — a model trained and tested within that cohort reaches r = 0.463 for iAUC. The drop to 0.227 is therefore attributable to **domain shift across cohorts**, not to noise in the external data.

The most uncomfortable result is the baseline comparison. On external iAUC and peak rise, the **carbohydrate-only Ridge regression transfers better** (r = 0.362 and 0.357) than the XGBoost model (0.227 and 0.223). The tree ensemble, which wins decisively inside CGMacros, appears to have fit cohort-specific interactions that do not survive the transfer, whereas the single, near-universal "more carbohydrate → larger excursion" relationship is more robust. The seven-feature variant adding fat and fibre changes little (external AUC r = 0.581, iAUC r = 0.272, n = 558). We treat this as a cautionary finding rather than a failure: it is exactly the kind of result that an external test is meant to expose, and it would be invisible under any within-cohort evaluation.

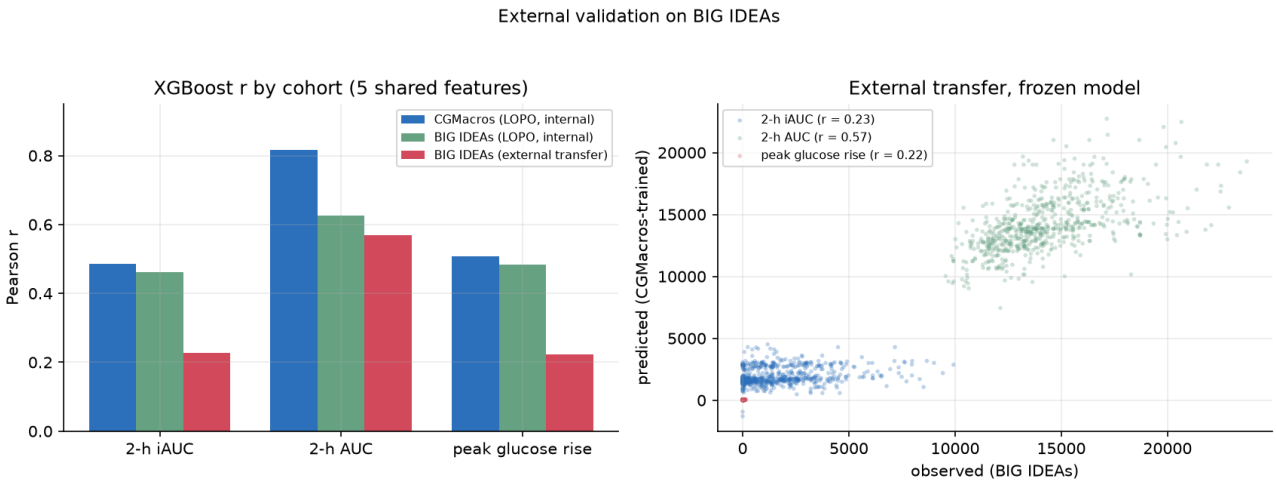


Figure 5 — External validation. Left: XGBoost Pearson r by cohort and target under the five shared features; the red bars are the frozen-model transfer. Right: predicted versus observed for the transferred model — the AUC cloud follows the diagonal loosely, while iAUC and peak rise are nearly flat.

4.5 Within- versus between-subject skill, and discrimination

A pooled correlation answers two questions at once, and it is worth separating them. **Within-subject** skill asks whether the model ranks a person's *own* meals correctly (each subject's observed and predicted values are centred on that subject's means before correlating). **Between-subject** skill asks whether the model recovers *who* tends to respond more (per-subject means are correlated across the 45 subjects). We also report a median-split ROC-AUC, which asks whether the model can pick out above-typical from below-typical responses.

Table 5 — XGBoost skill decomposition and discrimination (LOPO).

Subset	Target	Pooled r	Within-subject r	Between-subject r	ROC-AUC
Breakfast	2-h AUC	0.890	0.601	0.926	0.956
Breakfast	2-h iAUC	0.655	0.519	0.693	0.875
Breakfast	Peak rise	0.632	0.544	0.667	0.831
All meals	2-h AUC	0.838	0.654	0.921	0.910
All meals	2-h iAUC	0.451	0.430	0.501	0.744
All meals	Peak rise	0.542	0.455	0.697	0.770

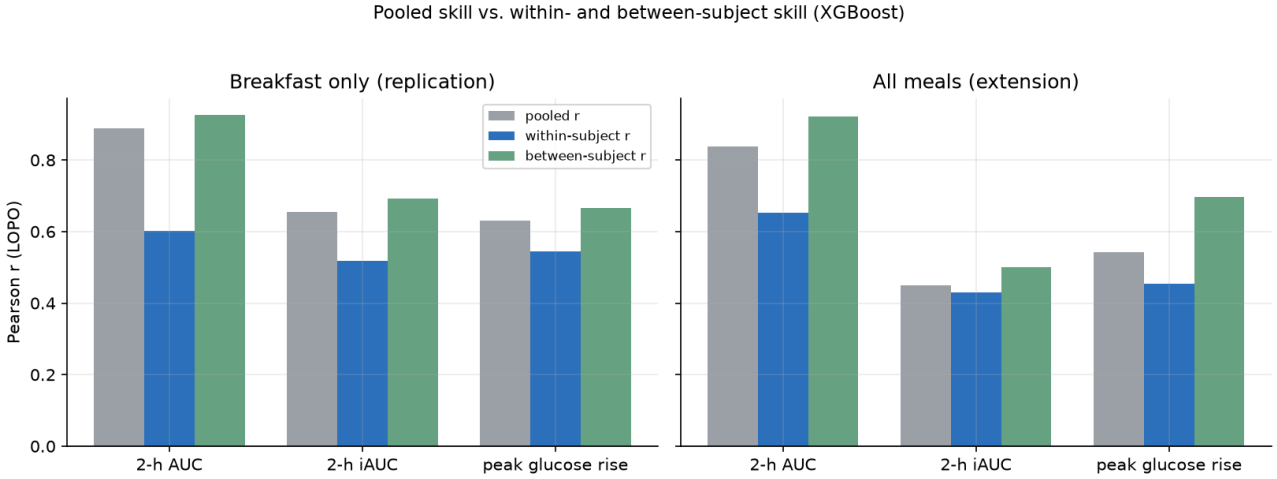


Figure 6 — Pooled skill versus its within- and between-subject parts (XGBoost, LOPO). When the between-subject bar towers over the within-subject bar, much of the pooled correlation reflects knowing *who* the person is.

The decomposition sharpens the AUC/iAUC contrast in §6. For AUC, between-subject skill (0.921, all meals) is far larger than within-subject skill (0.654): a substantial part of the impressive pooled AUC correlation is simply identifying individuals with a higher overall glucose level, not reading the meal. For iAUC, the two components are nearly equal (0.430 within, 0.501 between), so its correlation is much closer to a genuine meal-level signal — a more favourable picture of iAUC than the pooled number alone suggests, even though iAUC's absolute correlation is lower. Peak rise sits in between.

The ROC-AUC column shows that discrimination is strong on every target (0.74–0.96) and moves with the pooled r, so the modest iAUC correlation is not a threshold or scale artefact: the model genuinely ranks meals less reliably for the excursion than for the total area. For reference, the linear baselines reach within-subject r of only ≈0.21 for iAUC and peak rise, and the mean predictor has zero within-subject skill by construction; the full table, including each baseline's decomposition, is in `experiments/results.csv`.

4.6 Prediction intervals (conformal)

A point prediction is not enough for a health-adjacent task. We add **cross-conformal** intervals (Vovk et al.; Barber et al., 2021) on top of the existing LOPO predictions — no new model, and the point predictions are unchanged. For each held-out subject, the absolute residuals of all *other* subjects serve as a calibration set; the finite-sample conformal quantile `ceil((n+1)(1-alpha))` gives the half-width, and the interval is `ŷ ± q`. Nothing about a subject's own responses is used to size its interval.

Table 6 — Conformal intervals, XGBoost under LOPO.

Subset	Target	Nominal	Empirical coverage	Mean width (mg/dL·min)	Worst-subject coverage
All meals	2-h AUC	80%	0.800	6,119	0.18
All meals	2-h AUC	90%	0.898	8,751	0.40
All meals	2-h iAUC	80%	0.800	5,357	0.32
All meals	2-h iAUC	90%	0.897	7,909	0.50
Breakfast	2-h AUC	80%	0.801	6,986	0.00
Breakfast	2-h AUC	90%	0.898	9,808	0.00
Breakfast	2-h iAUC	80%	0.804	6,000	0.00
Breakfast	2-h iAUC	90%	0.896	8,885	0.30

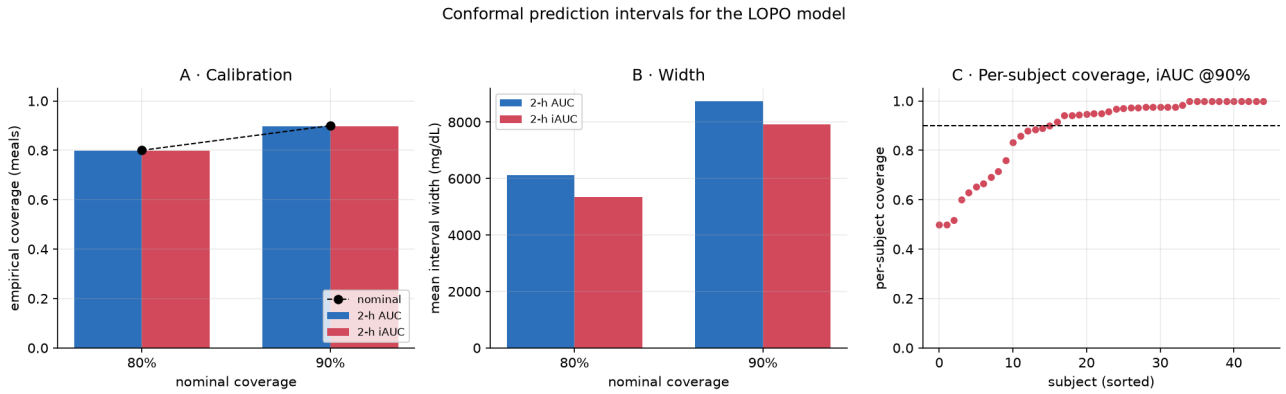


Figure 7 — Conformal intervals. A: empirical versus nominal coverage. B: interval width. C: per-subject coverage at 90% for iAUC, sorted — the tail below the dashed line is the exchangeability caveat made visible.

Three things are worth stating. First, **marginal coverage is essentially exact**: 0.800 and 0.898 against nominal 80% and 90%, as conformal theory predicts. Second, the intervals are **wide** — the 90% interval for all-meal iAUC spans ≈7,900 mg/dL·min, close to four times the median iAUC of ≈2,000, so the intervals are trustworthy but coarse; they shrink with the target's predictability (AUC intervals are narrower relative to its spread). Third, and most important, **coverage is not uniform across subjects**: at 90% the worst subject reaches 0.50 for all-meal iAUC and 0.30 for breakfast iAUC, but 0.40 for all-meal AUC and 0.00 for breakfast AUC; at 80% the worst breakfast subject has no meals covered for either target. Marginal validity can therefore hide individuals who sit systematically outside the band — a direct consequence of meals within a subject being correlated rather than exchangeable. Per-subject (Mondrian) or residual-normalised conformal methods are the natural remedy (§8).

4.7 Repeated meals: mixed-effects check

§4.5 gave a *predictive* split of the LOPO skill. Here we add the complementary *inferential* treatment of repeated meals (a mixed-effects model with a subject random intercept, plus within-subject centring and stratified within/between reporting). This model answers a different question and does not change the LOPO numbers.

For each target we fit a linear mixed model on 1,556 meals (one meal lacked a fibre value and was dropped) with a **subject random intercept** and the **Mundlak within-between** specification: each standardised meal-level predictor enters both as a within-subject deviation (`cw`, "this person ate more carbs than usual") and as the subject's mean (`cm`, "this person habitually eats more carbs"). The random intercept absorbs the repeated-measures structure.

Table 7 — Variance components and ICC.

Target	σ (subject)	σ (residual)	ICC (between-subject share)
2-h AUC	4,306.8	2,863.1	**0.694**
2-h iAUC	1,536.5	2,201.2	0.328
Peak rise	19.4	28.3	0.321

The ICCs are a direct, quantitative cross-check on §4.5. For AUC, about **69%** of the response variance sits *between* people; for iAUC and peak rise only about **32%** does. So AUC's pooled correlation being dominated by between-subject skill is not an artefact of the metric — it

reflects where the variance actually lives. It also matches the external estimate that repeated meals are only weakly reproducible within a person (Hengist et al., 2024; ICC 0.17–0.28).

Table 8 — Standardised within- vs between-subject effects (coefficient ± SE; only carbs and protein are shown — fat and fibre are indistinguishable from zero within subjects).

Target	Predictor	Within-subject	Between-subject
2-h AUC	carbs	**+581.6 ± 88.6**	−258.7 ± 3,337.5
2-h AUC	protein	**−340.2 ± 85.8**	+7,483.3 ± 4,913.8
2-h iAUC	carbs	**+605.6 ± 68.2**	+65.3 ± 1,221.9
2-h iAUC	protein	**−214.9 ± 66.0**	+3,701.8 ± 1,799.6
Peak rise	carbs	**+7.9 ± 0.9**	+0.3 ± 15.5
Peak rise	protein	**−2.9 ± 0.8**	+47.9 ± 22.8

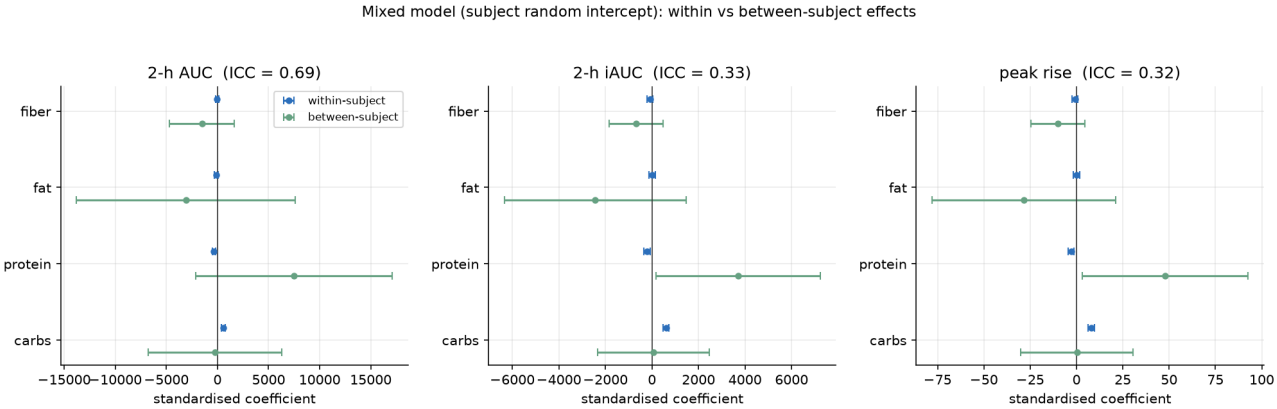


Figure 8 — Mixed model (subject random intercept). Standardised within- and between-subject coefficients with 95% intervals; each panel has its own x-scale because the targets differ by orders of magnitude.

Two conclusions follow. First, **within a person, carbohydrates raise and protein lowers the response**, significantly and in the same direction for all three targets — the two effects that survive once stable between-person differences are absorbed; fat and fibre do not. Second, the **between-subject coefficients are an order of magnitude less precise** (only 45 people) and, for protein, flip sign — the association across *people* points the opposite way to the one within a person. That divergence is a concrete reason to distrust population-level macronutrient predictors as personal ones, and it reinforces the caution in §6.

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5. Measurement-reliability ceiling analysis

The cross-cohort collapse in §4.4 and the published within-person repeatability figures (duplicate-meal r ≈ 0.45–0.46; §6) raise a question the preceding sections only gesture at: **how much of PPGR predictability is bounded by measurement itself, and is our model anywhere near that bound?** This section answers it with six targeted experiments, all produced by `src/ceiling_experiments.py` into `experiments/ceiling_results.csv`.

A methodological note first. CGMacros breakfasts repeatedly reproduce the same six macronutrient profiles and same-subject breakfasts match exactly, so they behave as a *repeated reference meal* — but this makes the within-cohort ceiling a *controlled-protocol*

ceiling, systematically friendlier than the inpatient duplicate meals behind the published 0.45–0.46 figures (Hengist et al., 2024, presented duplicates in 30 adults without diabetes; not free-living). We therefore keep the two regimes separate throughout and never compare across them. All six experiments were re-run on 2026-10-27 with the pinned suite in §9; every row outside E2 reproduced to four decimal places, and the E2 values below come from that run.

5.1 Within-cohort duplicate-meal reliability (E2)

Subset	Tolerance	Pairs (n)	Subjects	iAUC duplicate r [95% CI]	ICC(1,1)
Breakfast	±2 g	**174**	44	**0.799** [0.702, 0.867]	0.798
Breakfast (both iAUC > 0)	±2 g	169	44	0.810 [0.715, 0.876]	0.810
All meals (same type)	±2 g	291	44	0.746 [0.657, 0.807]	0.746
Dinner	±2 g	40	13	0.374 [0.088, 0.569]	0.357
Snack	±2 g	77	18	0.376 [0.069, 0.656]	0.376
Lunch	±2/±5 g	**0**	0	— (no duplicate pairs exist)	—
All meals (same type)	±5 g	455	44	0.685 [0.577, 0.757]	0.686

The 95% intervals resample *subjects* rather than pairs, because one subject can contribute several pairs and pair-level intervals would be too narrow.

The main E2 estimate uses all 1,699 meals, including non-positive responses, so it is **not selected on the outcome**; the prediction models, following the official protocol, use the `iauc > 0` filter. To make the two quantities directly comparable, the second breakfast row restricts pairs to meals with iAUC > 0 on both sides (169 of the 174 pairs). The filtered estimate is slightly higher (0.810), so the 0.799 headline is the conservative one for the model comparison in §5.1.

Correction note (2026-10-27). An earlier version of this table reported ICC ≈ 0.99. That implementation formed the between-pair mean square as `n × var(pair means)` instead of `2 × var(pair means, ddof=1)`, and the within-pair mean square from the two marginal variances instead of `mean((a−b)²)/2`; on a trivial input of perfectly agreeing duplicates it returned 0.5. The table now reports the correct one-way ICC(1,1) for k = 2, which lands close to r. The same run restricted the pooled "all meals" row to same-type pairs, as the code's docstring had always specified; the earlier cross-type value (0.715 for ±2 g, 363 pairs) is retained in `ceiling_results.csv` as `all meals (cross-type pairs)` for provenance.

Three facts matter here. First, the breakfast model's r = 0.655 (§4.2) sits **below** the duplicate-meal reliability of the same breakfasts (ρ = 0.799). Under the classical measurement model X = T + ε, a model that uses only stable information cannot correlate with a single observed meal above √ρ ≈ 0.894 (the parallel-measurement assumption); using ρ itself as the benchmark is therefore *conservative*. The model stays below both bounds — there is no "model beats measurement" paradox, and the published baseline (≈0.64) is consistent for the same reason. The 0.45–0.46 inpatient duplicate-meal figures describe a *noisier device/protocol regime*, not this one, and they are an optimistic benchmark for a free-living, self-reported target. Second, reliability is strongly meal-type-dependent: reference breakfasts are highly repeatable (0.80) while free-choice dinners and snacks are not (≈0.37–0.38). That pattern is consistent with protocol control rather than a fixed biological constant, and it means the breakfast ceiling cannot be transplanted to other meal types. Third,

CGMacros contains no duplicate lunches (n = 0), so any reliability claim in this cohort is necessarily breakfast-centric; we flag this as a protocol limitation rather than paper over it.

5.2 Is iAUC skill riding on baseline glucose? (E5)

Because AUC ≈ 120 × baseline + iAUC by construction, a model evaluated on AUC can score well while encoding little about the *excursion*. The ablation quantifies this (LOPO, XGBoost, 19→18→1 features):

Target	Subset	Full (19)	No baseline (18)	Baseline only (1)
iAUC	all meals, iauc>0 (n=1,557)	0.451	**0.457**	0.204
iAUC	breakfast (n=423)	0.655	**0.657**	0.542
AUC	all meals, iauc>0	0.838	0.638	**0.807**
AUC	breakfast	0.890	0.738	**0.861**
Peak rise	breakfast	0.632	0.647	0.547

(Unfiltered, n=1,699, agrees: iAUC 0.446 / 0.432 / 0.123; AUC 0.807 / 0.636 / 0.751.) Two conclusions. **iAUC skill is invariant to removing baseline glucose** — dropping it changes r by less than 0.01 — so the headline iAUC numbers are not circular. **AUC, in contrast, is largely a baseline-glucose shadow**: baseline alone recovers r = 0.81–0.86, and removing baseline costs 0.15–0.20. This is the quantitative basis for treating iAUC, not AUC, as the honest target throughout this report.

A sharper reading of the breakfast row: baseline glucose alone reaches 0.542 on breakfast iAUC, so the *marginal* contribution of all 18 meal and blood features is ≈ 0.11 (0.655 − 0.542). Most of what looks like "meal response prediction" at breakfast is, on inspection, fasting-state information.

5.3 Who responds vs how much: person-level aggregation (P3)

Averaging predicted and observed values within each subject (n = 45 persons) asks a different question — *can the model rank people by their typical response?* — and the answer is more forgiving than the meal-level one:

Target	Meal-level r	Person-mean r
iAUC (all meals)	0.451	**0.501**
AUC (all meals)	0.838	**0.921**
Peak rise (all meals)	0.542	**0.697**
iAUC (breakfast)	0.655	0.693

The most striking number is what baseline glucose alone does at the person level: **r = 0.963 for person-mean AUC** (0.71–0.76 for iAUC, exceeding the full-feature model's 0.50 — with only 45 points we read this as noise around a genuinely strong baseline signal, not as a feature-selection finding). "Who responds more" is almost fully determined by fasting state; the model's person-level edge beyond baseline is real but modest. This sharpens the within-/between decomposition of §4.5 and §4.7 from a variance statement into a predictive one.

5.4 Bidirectional external transfer (E3)

§4.4 transfers CGMacros → BIG IDEAs. The reverse direction completes the picture (frozen BIG IDEAs LOPO model applied to CGMacros, 5 shared features):

Target	CG → BI (forward)	BI → CG (reverse)
iAUC	0.227	**0.085**
AUC	0.569	0.748
Peak rise	0.223	0.174

iAUC transfer fails in both directions (0.227 / 0.085). This rules out the most convenient explanation of the §4.4 collapse — that 16 subjects are simply too few to evaluate on — because the reverse direction evaluates on 1,699 meals from 45 subjects and fails harder. AUC transfers partially both ways, but §5.2 already showed AUC is baseline-driven: what "transfers" is the compatibility of baseline-glucose distributions, not meal-response skill.

5.5 Does person-level aggregation rescue cross-cohort transfer? (E4)

Since aggregation helped within-cohort (§5.3), we test whether ranking the 16 BIG IDEAs subjects by person-mean predictions recovers signal:

Target	Meal-level r (n=656)	Person-mean r (n=16)	95% CI (person)
iAUC	0.227	**−0.029**	[−0.43, 0.43]
AUC	0.569	**0.715**	[0.35, 0.90]
Peak rise	0.223	0.054	[−0.33, 0.44]

Aggregation lifts cross-cohort AUC (0.569 → 0.715) — again consistent with baseline dominance — but does **not** rescue cross-cohort iAUC, which is statistically indistinguishable from zero at n = 16. Within-cohort aggregation gains do not survive the domain gap.

5.6 Synthesis

The six experiments form a single chain:

1. Inside CGMacros, the breakfast model (0.655) stays below the standardized-breakfast duplicate-meal reliability (ρ = 0.799), and below the corresponding √ρ ≈ 0.894 bound for a model that uses stable information only — no paradox, published baseline included (§5.1).
2. iAUC skill is not baseline circularity; AUC is (§5.2).
3. Person-level aggregation sharpens within-cohort rankings, and person-mean AUC is nearly fully determined by baseline glucose (§5.3).
4. iAUC transfer fails bidirectionally, so the cross-cohort collapse is a property of the cohort/device/annotation gap, not of sample size (§5.4).
5. The closest available device-level reliability (~0.45–0.46, inpatient duplicate meals; an optimistic benchmark for a free-living target) is the reference on the external side, and the transferred model (0.227) sits below it — while within-cohort skill sits below *its* (higher) ceiling. The target regime's own ρ is unmeasured, which is itself part of the argument.

The paper-level claim this supports is deliberately narrow: **published PPGR model performance must be read against the measurement reliability of its own evaluation regime, and cross-regime comparisons — including our own §4.4 transfer — are confounded by differences in that reliability and cannot be read as comparisons of model quality. Reliability explains part of the gap; the residual reflects model, feature and cohort differences that reliability alone cannot capture.**

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6. Discussion

Four findings are worth stating plainly.

First, **the published baseline reproduces**, and the extension is credible. Matching r ≈ 0.89 (AUC) and r ≈ 0.64 (iAUC) on breakfast means subsequent numbers sit on a validated pipeline rather than an idiosyncratic one.

Second, **non-linearity earns its place**. The best linear model reaches r = 0.255 on breakfast iAUC, whereas the boosted stumps reach 0.655. With only 80 depth-1 trees, XGBoost is still a small model; the gain comes from modelling thresholds and interactions (for example, the effect of carbohydrate differing by baseline glucose), which a single linear coefficient cannot express.

Third, **AUC is easier than iAUC**, consistently. AUC is dominated by the subject's overall glucose level, which the baseline and HbA1c features capture well. iAUC subtracts that baseline and isolates the *excursion*, the part that depends on the meal and on individual handling of it — and that is harder. This distinction is often blurred in PPGR reporting, where a high AUC correlation can be mistaken for good meal-response modelling. The within-/between-subject decomposition in §4.5 makes the point directly: AUC's pooled r is dominated by between-subject skill, whereas iAUC's two components are nearly equal.

Fourth, **external validation changes the story in a useful way**. Inside CGMacros, XGBoost dominates the linear baselines on every target. Across cohorts, the ranking partly reverses: the tree model still transfers for AUC (r = 0.569) but loses to a one-variable carbohydrate regression on iAUC and peak rise. The lesson is not that gradient boosting is wrong — it is that a model whose advantage comes from cohort-specific interactions carries a hidden generalisation risk, and only an external test reveals it. We would rather report this than present a single-cohort number that flatters the method.

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7. Limitations

We state these explicitly; they bound every claim above.

- **Small cohort.** 45 subjects cannot support deep models or fine-grained subgroup analysis. Neural approaches that need tens of thousands of samples are not justified here, which is why we use gradient-boosted stumps.
- **Pooled metrics mix two variance sources.** The headline correlations are computed over pooled meals, so between-subject differences inflate them relative to a purely within-subject evaluation — for AUC, most of the pooled value is between-subject skill

(§4.5). We report the decomposition alongside the pooled metric, but the pooled number alone should not be read as meal-level skill.

- **The response is only weakly repeatable within a person.** The same meal, eaten again a week later, does not reliably reproduce its excursion: reported within-subject ICCs for postprandial response are only 0.17–0.28 (Hengist et al., 2024), with duplicate-meal Pearson correlations of r = 0.46 (Abbott) and r = 0.45 (Dexcom). Our own mixed model agrees, with ICC 0.32 for iAUC and peak rise (§4.7); for AUC the ICC is 0.69, i.e. most of that variance is between people rather than between meals of the same person. This bounds how strongly *any* stable-information composition model can correlate with a single measured meal: under a parallel-measurement model the bound is $\sqrt{\rho}$ ($\approx$0.67 for ρ = 0.46), with ρ itself the conservative benchmark. It is a property of the cohort we use, not of our pipeline.
- **CGM measurement is noisy, lagged and device-dependent.** Subcutaneous glucose lags blood glucose by roughly 9–10 min, which blurs fast postprandial peaks; because our window is anchored on the annotated meal time, the lag shifts the target rather than invalidating it. Measurement also depends on which sensor channel is used, and the two channels in this study agree only moderately on meal ranking ($\tau \approx 0.43$; see §2.2 and Howard et al., 2020).
- **Confounding is uncontrolled.** Meal timing, the interval since the previous meal, physical activity, sleep and menstrual cycle all influence the postprandial response and are not features here. The correlations above therefore describe association under this study's conditions, not an isolated causal effect of macronutrients.
- **Meal annotation quality differs across cohorts.** CGMacros meals are study-annotated; BIG IDEAs is a free-living, self-reported food log subject to the usual portion-estimation error. Part of the cross-cohort drop in §4.4 is therefore a difference in measurement, not only in modelling.
- **External validation is narrow and imperfect.** We test one external cohort (BIG IDEAs, 16 subjects, 656 meals) with only five shared features, because that cohort lacks the CGMacros blood panel. The two studies differ simultaneously in participants, CGM device (Libre Pro vs Dexcom G6) and meal annotation (study-annotated vs free-living self-report), so the drop in performance cannot be attributed to any single factor. The BIG IDEAs meals are also derived from a 1.1.2 food log whose dates required our own repair procedure, which is validated but not ground truth.
- **Intervals are only marginally valid.** We do report conformal intervals (§4.6), and their marginal coverage matches nominal (0.800 / 0.898). But the coverage is not uniform across people: at 90% the worst subject reaches 0.30 for breakfast iAUC and 0.50 for all-meal iAUC, while three of the four breakfast cells (AUC at 80% and 90%, iAUC at 80%) have a worst subject at zero. The intervals are also wide. They should be read as population-level guarantees, not per-person ones.
- **Observational and non-causal.** Associations between features and response are not interventions. Nothing here implies that changing a macronutrient will cause the predicted change.
- **Reproducibility in this field is poor, which cuts both ways.** An audit of 67 deep-learning glucose-prediction papers (Lu et al., 2026) found that only 23.9 % of the studies released code, 37.3 % used private data, and 55.2 % of all 67 relied on OhioT1DM — a 12-person cohort. That makes an open, subject-wise-evaluated pipeline

a genuine differentiator, but it also means most published numbers are not directly comparable with ours.
- **Data provenance caveats.** The dataset's internal `LICENSE.txt` is an empty file; the CC BY-NC-SA 4.0 terms are declared only on the PhysioNet web page. We treat the web page as authoritative and comply with attribution and share-alike.

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8. Future work

1. **More external cohorts and domain adaptation**, building on the BIG IDEAs result. Re-calibration or a small amount of target-cohort data may recover the iAUC signal that the frozen model loses; testing a third cohort (e.g. a T2D population) would separate device effects from population effects.
2. **Adaptive / per-subject conformal prediction** (Mondrian or residual-normalised), so interval coverage is uniform across people rather than only marginal — §4.6 shows the current intervals under-cover specific individuals even though the pooled coverage is on target.
3. **Richer meal representations** — meal timing, prior-meal context, activity, and **meal type as an explicit feature** (we currently use meal type only to define evaluation subsets, not as an input to the model).
4. **Per-subject reporting on larger cohorts.** The within-/between-subject split in §4.5 is pooled over 45 subjects; per-subject correlations with confidence intervals need more meals per person than this cohort provides.

An interactive demo exposing per-meal predictions and exact TreeSHAP attributions on this frozen pipeline is already deployed (https://metanutri-ai-ppgr-predictor.streamlit.app/); see §9 for the code.

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9. Reproducibility

All results are produced by a small, dependency-pinned pipeline (`research/`), independent of the MetaNutri platform. The entire chain — from raw download to the figures and this PDF — runs with **one command** from `research/`:

```
bash src/run_all.sh                    # full pipeline
bash src/run_all.sh --skip-download    # reuse already-downloaded archives
```

It runs, in order:

```
src/download_data.sh      # parallel-chunk download + SHA256 verification
src/download_bigideas.py # fetch the 33 small BIG IDEAs files (2.4 MB)
src/build_dataset.py      # CGM → meal-level table (uses src/iauc.py for targets)
src/experiment.py         # LOPO protocol, baselines, XGBoost (uses src/evaluate.py);
                          #   breakfast replication × all-meals extension × three targets
                          #   → experiments/results.csv
src/build_external.py     # BIG IDEAs → meal-level table (date-offset repair)
src/external_validate.py # cross-cohort transfer → experiments/external_results.csv
src/make_figures.py       # figures 1–6
src/uncertainty.py        # conformal intervals → table 6, figure 7, experiments/
uncertainty.csv
src/mixed_effects.py      # subject random intercept → tables 7–8, figure 8
                          #   → experiments/mixed_effects_{variance,coefficients}.csv
(statsmodels)
src/ceiling_experiments.py # reliability-ceiling analyses (§5): duplicate-meal
reliability,
                          #   baseline ablation, person-level aggregation, bidirectional
                          #   transfer → experiments/ceiling_results.csv
src/train_final.py        # freeze the model into app/model/ for the demo
src/export_report_pdf.py # this report → reports/technical_report.pdf
```

(`iauc.py` and `evaluate.py` are library modules imported by the steps above, not separate entry points.)

Pinned versions: `numpy==2.5.3`, `pandas==3.0.6`, `scikit-learn==1.9.1`, `xgboost==3.4.1`, `scipy==1.18.1`, `matplotlib==3.11.2`, `statsmodels==0.15.0`. The original runs used Python 3.12; the 2026-10-27 re-run of the ceiling experiments used Python 3.14.8 and reproduced every non-E2 row to four decimals. Random seed 42. Re-running the suite regenerates `experiments/results.csv`, `experiments/uncertainty.csv` and `experiments/mixed_effects_*.csv`, `experiments/ceiling_results.csv`, and `make_figures.py` regenerates the figures.

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10. Data attribution and license

The full license layering for this module (code vs. data-derived artifacts vs. datasets) is specified in `research/NOTICE.md`; the summary below is part of that layering.

This work uses the **CGMacros** dataset:

- Source: PhysioNet — https://physionet.org/content/cgmacros/1.0.0/
- DOI: `10.13026/3z8q-x658`
- Reference: Das et al., *Scientific Data* 12, 1557 (2025).
- License: **Creative Commons Attribution–NonCommercial–ShareAlike 4.0 (CC BY-NC-SA 4.0)**.

Use is non-commercial (research and portfolio). This analysis is a derivative work and is distributed under the same license, with attribution to the original authors. Raw data is never redistributed in this repository.

External validation uses the **BIG IDEAs Lab Glycemic Variability and Wearable Device Data** dataset:

- Source: PhysioNet — https://physionet.org/content/big-ideas-glycemic-wearable/1.1.2/
- DOI (version 1.1.2, used here): `10.13026/zthx-5212`

- Reference: Cho, P., Kim, J., Bent, B., & Dunn, J., *BIG IDEAs Lab Glycemic Variability and Wearable Device Data*, PhysioNet.
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